



Optimal design of a hybrid energy system considering techno-economic factors for off-grid electrification in remote areas

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Abstract

Somalia encounters several concerns involving widespread power outages and high reliance on imported fossil fuels. Nonetheless, renewable energy can viably meet the escalating energy demand in Somalia. This study investigates the techno-economic feasibility and optimal design of hybrid solar photovoltaic (PV), diesel generator (DG), and battery energy storage systems (BESS) in the remote areas of the Lower Shabelle region in Somalia. A dependable, sustainable, and inexpensive electricity source with low environmental pollution emissions was determined. The technical formation and economic feasibility of the constructed power system were developed using HOMER Pro® software. Three hybrid energy systems in the analysis were compared, including PV, DG, and BESS. The system performance and economic feasibility depended on numerous parameters, such as load fulfillment, net present cost (NPC), levelized cost of energy (LCOE), excess electricity, operation and maintenance costs, renewable fraction, fuel cost, payback time, and CO₂ emissions. A comparison analysis demonstrated that the PV/DG/BESS hybrid system was more dependable, economically viable, and suitable for rural electrification than other investigated configurations (PV/BESS and DG/PV systems). The PV/DG/BESS hybrid system resulted in an NPC of \$129,679 and an LCOE of \$0.180/kWh. The payback period, return on investment, and internal rate of return were 1.8 years, 65%, and 66.6%, respectively. Approximately 2.27% of excess electricity and 12,413 kg/year of environmental pollutant emissions were recorded. Consequently, this sustainable system could be utilized in isolated regions worldwide. The outcome

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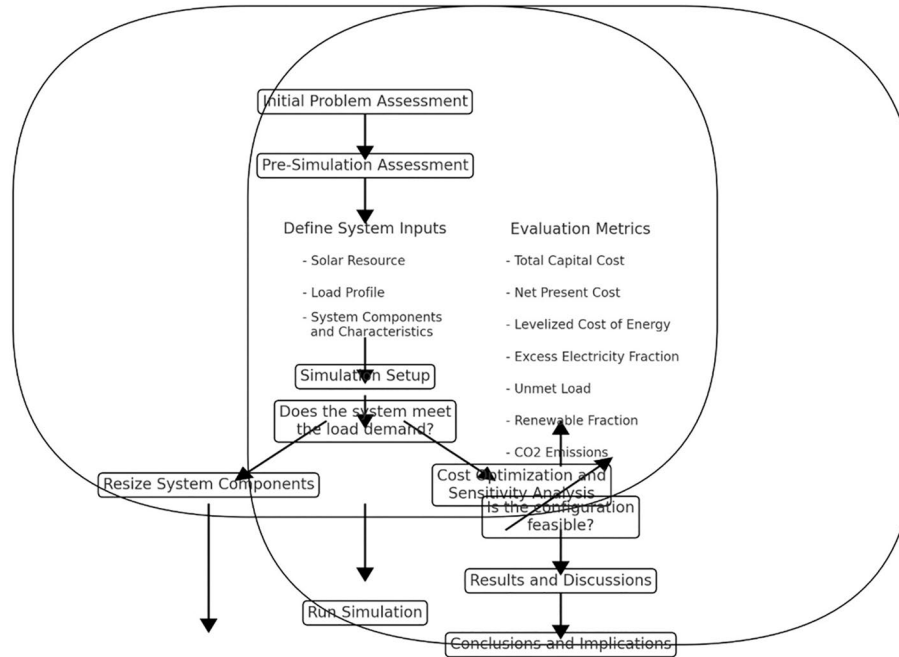
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of this study paralleled the global strategic targets outlined in Sustainable Development Goal 7 (SDG7), “Affordable and Clean Energy”, of the United Nations.

Graphical abstract



Keywords Techno-economic feasibility · Hybrid energy system · Renewable energy · Levelized cost of energy · Excess electricity · Somalia

Abbreviations

AC	Alternating current	LCOE	Levelized cost of energy
BECO	Banadir electric company	LF	Load following
BESS	Battery energy storage system	NPC	Net present cost
BE	Baseline emission	NREL	National renewables energy laboratory
CC	Cycle charging	PE	Project emission
CO ₂	Carbon dioxide	PV	Photovoltaic
COE	Cost of energy	RESs	Renewable energy sources
CRF	Capital recovery factor	REP	Renewable energy projects
CSF	Capacity shortage fraction	RF	Renewable fraction
DC	Direct Current	SDGs	Sustainable development goals
DG	Diesel Generator	SOC	State of charge
EEF	Excess electricity fraction	$C_{ann,tot}$	Total annualized cost per year (\$/year)
EMS	Energy management strategy	C_{rep}	Replacement cost
ESS	Energy storage system	E_{pro}	Annual electrical production (kWh/year)
HOMER	Hybrid optimization of multiple energy resources	E_{excess}	Total annual excess of electricity (kWh/year)
HRES	Hybrid renewable energy systems	EE_{grid}	Grid emission factor
HESs	Hybrid energy systems	$E_{non-ren}$	Non-renewable electrical (kWh/year)
IEA	International Energy Agency	$E_{required}$	Annual electricity (kWh/year)
kW	Kilo watt	E_{served}	Electrical load served (kWh/year)
kWh/m ² /day	Kilo watt hour per meter square per day	F_{PV}	PV derating factor (%)
kWh	Kilowatt hour	$F_{0,gen}$	Coefficient of the generator fuel curve

$F_{l,gen}$	The slope of the fuel curve for the generator
$G_{T,NOCT}$	Solar radiation at nominal operating cell temperature
$H_{non-ren}$	Renewable thermal (kWh/year)
H_{served}	Total thermal load served (kWh/year)
I_{max}	Maximum charging current of the battery (A)
T_a	Air temperature
$T_{C,NOCT}$	Nominal operating cell temperature
$T_{a,NOCT}$	Ambient air temperature at NOCT
T_C	Cell temperature ($^{\circ}\text{C}$)
T_{comp}	Component lifetime
$T_{c,STC}$	Cell temperature under standard test conditions (25°C)
T_P	Project duration
$T\alpha$	Temperature coefficient of power ($\%/^{\circ}\text{C}$).
P_{PV}	PV array power output (kW)
P_D	Total power demand
P_{DG}	Total power generated
V_{nom}	Nominal voltage of the battery
Y_{PV}	PV power output at STC (kW)
$\eta_{Batt, C}$	The efficiency of battery charge storage
η_{cnv}	Converter efficiency
α_P	Temperature coefficient of power ($\%/^{\circ}\text{C}$)
α_c	Battery maximum charge rate (A/Ah)

Introduction

Energy crisis mitigation is concerning in the twenty-first century, where expensive and unreliable power access is a significant obstacle for developing countries (S. Li et al. 2022a, b, c). This issue results in energy poverty and

global inequality, with approximately 17% of the world-wide population requiring electricity access (Olatomiwa et al. 2018). Approximately 85% of the population resides in the remote rural parts of sub-Saharan Africa (SSA) and several emerging nations. Notably, the SSA population generates the highest energy deficiency rate, in which 42% of the population acquires access to electricity (Kinally et al. 2022). Given the significance of energy access in modern civilization, the Sustainable Development Goals (SDGs or Global Goals) serve as a road map for constructing a more sustainable future for all populations (Bose and Khan 2022). Affordable and clean energy is integral to attaining the other SDGs established by the United Nations (UN), specifically SDG 11 (Sustainable Cities and Communities) (Bruce and Vinuales 2021; Fs et al. 2021). Figure 1 depicts the SDG examples relevant to this study.

The SDGs are a collection of global objectives with an integrated strategy to solve worldwide social, economic, and environmental concerns. These concerns are addressed by providing cheap and clean energy, boosting economic growth, and protecting the environment for future generations (Sánchez et al. 2022). Nonetheless, the power shortage negatively impacts the residents' quality of life, health, stability, and economic activities in isolated rural areas far from the primary grid (Olatomiwa et al. 2018). Therefore, electricity access decreases poverty through human living standards, employment rate, and productivity improvements (Asghar et al. 2022). Renewable energy sources (RESs) are crucial to developing eco-friendly and sustainable power systems that successfully reduce fossil fuel reliance and environmental damage (David et al. 2024; Ahmed et al. 2022; Nham and Ha 2023). Notable RES examples include wind, small hydropower, and solar PVs.



Fig. 1 Diagram indicating the SDGs relevant to this study

Although renewable energy (RE) power generation on a stand-alone basis is a viable alternative for rural communities, it is not practical in primary national grids (Ocampo et al. 2021). The primary issues connected with RE sources are unpredictability and variability, which impact the quality and reliability of the energy infrastructure (Wankou Ngouleu et al. 2023). One of the most promising approaches to the issues mentioned above is the use of energy storage systems (ESSs), which offer a long-term way to balance supply and demand (Esmailion et al. 2024; Esmailion and Soltani 2024). ESSs play a crucial role in enhancing grid stability, optimizing energy system performance, enabling voltage frequency regulation, and mitigating the environmental impact of fossil fuel consumption across diverse power system applications (Al-Shetwi et al. 2020; Jafarizadeh, Soltani, and Nathwani 2020).

Somalia mainly relies on fossil fuels for power generation and transportation. Additionally, charcoal is extensively used for cooking, while imported oils serve various other purposes. Fossil fuels account for 87.8% of electricity generation, while RE contributes 12.2% (Samatar et al. 2023). Nevertheless, oil imports deprive the country of hard currency, while charcoal extraction from firewood promotes deforestation. Most regions generate energy with diesel generators (DGs) due to the lack of a national power grid in Somalia, while substantial costs are related to the operation and upkeep of the DGs (Samatar et al. 2023). Currently, Somalia requires over 121,000 L of diesel fuel daily to sustain the country's generation capacity following wet-stacking conditions (Samatar et al. 2023). Rapid urbanization can also increase power generation and the number of users. Hence, daily diesel consumption could achieve 694,000 L by 2024 (Technical, Facility, and Sector 2022; MOP 2020). As most electricity is produced by local Independent Power Producers (IPPs), generating and distributing electricity in a free-market setting is a significant economic policy concern, with users and businesses paying between \$0.3 and 1/kWh (ILO 2016).

The World Bank data reveals that Somalia has a low electricity access rate, with the current national electricity access rate at 49%, and approximately 9 million Somalis still lack access to electricity (UNSD, WHO, IEA, IRENA 2021). Recent SDG7 (Affordable and Clean Energy) data suggests that the urban and rural electricity access rates in Somalia are 71 and 31%, respectively (UNSD, WHO, IEA, IRENA 2021). Therefore, the government established a National Regulatory Authority in 2021 following the 8th (2017–2019) and 9th (2020–2024) National Development Plans (NDPs). These plans aim to achieve market reforms and improve energy access between 15 and 45% by 2024 (PLAN and 2019 2016; Planning 2020). The Ministry of Planning also formed an investment promotion bureau [Somalia Investment (SOMINVEST)] to attract foreign

direct and energy investments as one of its priority industries in 2015. As many rural and distant communities rely on diesel, raising the RE proportion is essential for realizing the government's energy policy objectives. This requirement minimizes fuel consumption and pollutants, thereby reducing the cost of energy (COE). The Federal Government of Somalia approved the implementation of RE-based off-grid electrification for remote rural communities in 2021. Thus, this study contributed to the goal of electrifying Somalian rural areas by proposing hybrid integrated energy systems. The study outcome aided numerous rural residents, which were subjected to inefficient and unreliable power infrastructures in Somalia.

Worldwide off-grid and hybrid energy systems (HESs) have been significantly investigated for isolated rural electrification. Thirunavukkarasu and Sawle studied the off-grid solar/wind/diesel/battery system in India, which was the most economical in remote locations (Thirunavukkarasu and Sawle 2021). Another study by Fodhil et al. evaluated the HESs in Algeria and recommended that PV/diesel/battery could produce 93% RE using particle swarm optimization (PSO). The study described the system as more economically feasible than PV/diesel alone (Fodhil et al. 2019). Meanwhile, Benchaabane et al. investigated the wind/diesel system using a storage system in Canada (Benchaabane et al. 2019). The study discovered that the wind/diesel power system paired with a storage system was optimal for meeting the electrical demands of remote places. Alternatively, Das et al. reported that the PV/diesel/pump-hydro system was significantly more cost-efficient than the PV/diesel/battery-based system in rural areas of Bangladesh (Das et al. 2021).

A study conducted by Alonso et al. revealed that optimizing renewable and diesel-hybrid mini-grids in Nyabikehe camp, Rwanda, might reduce costs and emissions by 32% and 83%, respectively (Baranda Alonso, Sandwell, and Nelson 2021). Another study by Halabi et al. evaluated two distributed power plants in Sabah, Malaysia, that possess unique PV/diesel system converters and storage battery combinations (Halabi et al. 2017). Similarly, Cai et al. examined PV/diesel off-grid systems in South Khorasan, Iran (Cai et al. 2020). The study demonstrated that integrating solar energy with a DG lowered carbon dioxide emissions by 59.6% and supply costs by 22.2%, more than a single diesel system. In Bangladesh, Iskanderani et al. explored an off-grid hybrid PV/diesel/battery system (Iskanderani et al. 2020). Although the suggested method could not reduce energy prices or overall expenses, it could still assist in reducing diesel dependence.

Agyekum and Nutakor assessed the feasibility of two power systems in southern Ghana, which evaluated the technological and economic potentials of hybrid PV/wind/diesel/battery and wind/diesel/battery systems (Agyekum and Nutakor 2020). The study suggested that both systems

were advantageous and produced electricity from multiple sources. Meanwhile, Radley and Lehmann-Grube examined 96 research publications and discovered that increasing off-grid solar power positively affected economic growth (Radley and Lehmann-Grube 2022). Other hybrid system designs were also proposed, with numerous studies evaluating their techno-economic feasibility and applicability in remote locations (Masrur et al. 2020; Murugaperumal and Ajay D Vimal Raj 2019). Furthermore, several researchers addressed the size and optimization of HESs in stand-alone settings and the competitive assessment regarding demand-side management (Das et al. 2017; Ma et al. 2018). Table 1 tabulates several HESs with different efficiency evaluation parameters.

Several studies have revealed that a combination of renewable energy (RE) sources, distributed generators (DGs), and battery storage technologies mitigates off-grid instability and provides electricity to remote and off-grid locations. In contrast, the technological and financial viability of hybrid renewable energy systems (HRESs) is not guaranteed due to variances in RE potential and meteorological data across different locations. Thus, the viability of the system necessitates assessment through the analysis of numerous system configurations at each site. However, earlier studies emphasized stand-alone systems without considering the excess electricity and environmental impact of PV systems. Since local energy sources and RE remain abundant in isolated regions, remote areas can establish the social viability of RE resources (Jain and Sawle 2021; Gambino et al. 2019; C. Li et al. 2022a, b, c). Consequently, this resource promotes small and large projects in underdeveloped nations, including Somalia. Combining energy sources also increases electricity generation efficiency (Dadak et al. 2022; Mehiri, Berkouk, and Mekhilef 2022; Rawa et al. 2021; Mokhtara et al. 2021), while off-grid HESs are practical and significantly investigated in several cities worldwide (Javed et al. 2021; Sawle et al. 2018). A techno-economic feasibility study should meticulously assess the merits and demerits to determine the overall viability and sustainability of implementing a hybrid energy system (HES) for off-grid electrification in a semiarid climate village. Therefore, it is crucial to consider the location's unique characteristics, the community's specific energy needs, and the available resources. This consideration is vital for making informed decisions that align with the community's long-term goals and broader sustainability objectives.

After conducting an extensive review, the authors have found that no study has been conducted on HESs in Somalia. Nonetheless, there is a noticeable knowledge gap regarding the techno-economic and optimal design of a PV/DG/BESS hybrid system in the country (Breyer et al. 2022). This study addresses the knowledge gap that has impeded Somalia's economic and sustainable development. Given the lack of consensus on the optimal sizing and feasibility

of power systems in Somalia's remote and urban areas, a definitive solution remains doubtful. This paper presents a case study of Somalia, investigating the technical and economic viability of implementing HESs in these communities. Due to the impracticality of grid extension in this area, the exploration of integrating RE source with an existing DG system is considered as the optimal solution. Therefore, numerous critical research gaps are characterized based on the following literature reviews: (a) a literature review of the renewable fraction (RF) economic conditions did not contain excess electricity fraction limits. Moreover, the optimal HES design was not described to examine the link between RF and excess power, (b) only a few studies merged the steady power output of DGs with the variable power generation from PV sources. These studies did not investigate the energy management strategy of HESs in remote locations and during off-grid operations, (c) no prior research has been conducted on the techno-economic feasibility design and comparative assessment of various hybrid RE system combinations in the Somali climate, including PV/BESS, PV/DG, and PV/DG/PV, and (d) the most studies have overlooked the environmental impact of both grid-connected or off-grid of PV systems. This study aims to address these challenges by conducting a comprehensive analysis focused on the techno-economic aspects and identifying the optimal system for HESs within the context of sustainable rural electrification in semiarid climate regions. Given these challenges, this paper bridged the research gaps and expanded the current body of knowledge through the following major contributions:

- 1) A comprehensive framework and techno-economic viability of accessible energy resources is proposed.
- 2) A HES was introduced, using a load-following dispatch technique to ensure sustainable rural electrification with stable and long-term electricity provision.
- 3) A novel off-grid HES maximized size and cost-effectiveness while convincing the optimal load demand with excess power, preserving minimum net present cost (NPC) and levelized energy cost (LCOE) levels, and keeping excess electricity below the 5% standard.
- 4) This study considers the environmental impact of a stand-alone PV system.
- 5) A comprehensive sensitivity analysis is performed to assess the system's viability across different scenarios.
- 6) The outcome guides and provides information for formulating and planning a rural electrification framework based on the energy system in the Somali power sector, thereby aiding the government and key stakeholders.

The novelty of our study lies in addressing HESs that have received limited attention in previous research efforts. While most studies have touched upon this topic, they have

Table 1 Review summary of HES-based optimization studies

Ref	System components	Method	System	Site	Research gaps
(Berbaoui et al. 2020)	PV/diesel/wind/ battery	HOMER and MATLAB	Off-grid	South Algeria	Both economic and environmental aspects are presented. However, the obtained results reveal optimal performance. The excess electricity is not considered
(Raghuwanshi and Arya 2020)	PV/diesel/ battery	HOMER	Off-grid	India	The paper discusses financial and environmental sustainability, excluding considerations related to excess electricity
(Nejabatkhah 2018)	PV/diesel/ battery	MATLAB	Off-grid	Canada	A hybrid PV-diesel-BESS microgrid has been developed without considering the PV system's excess electricity and environmental impact
(Halabi and Mekhilef 2018)	PV/diesel/ battery	HOMER	Off-grid/ grid-connected	Malaysia	This paper does not address the environmental impacts of the PV system
(Wu et al. 2021)	PV/diesel/ battery	MATLAB	Off-grid	China	The microgrid system's reliability assessment lacks consideration for excess electricity and its environmental impact
(Cai et al. 2020)	PV/diesel/ battery	GIS	Off-grid	Iran	Economic and environmental considerations, including DG, are considered, but the environmental impacts of PV and excess electricity are not adequately explored
(Allouhi and Rehman 2023)	PV/wind/ battery	HOMER grid	Grid-connected	Morocco	The optimal design is proposed; however, the PV system's excess electricity and environmental impacts are not addressed
(Bedadi and GebreMichael 2021)	Hydro/PV/ battery	HOMER Pro	Off-grid	Rwanda	The analysis covers techno-economic aspects but neglects excess electricity and the environmental impact of the PV system
(Tiam Kapen et al. 2022)	PV/FC/Electrolyzer/BG & PV/BESS/FC/Electrolyzer/BG	HOMER Pro	Off-grid	Cameroon	This study presents the techno-economic feasibility, excluding environmental aspects and excess electricity considerations

lacked tangible facts regarding the excess electricity and environmental impact of PV systems, thereby failing to underscore its significance. In our investigation, we initially optimized a foundational system for conducting simulations using HOMER Pro® software. Subsequently, we propose and develop a novel system based on this optimized framework to explore the potential for excess electricity using the same software. Through meticulous analysis of existing data, we present a more concise and comprehensive model of the energy system.

The remainder of this paper is outlined: “[Methodology](#)” Section provides the materials, site data, and research techniques. “[Mathematical modeling of the system component](#)” Section explains the proposed hybrid power system model, while “[Results and discussions](#)” Section presents results and discussions. Finally, “[Conclusions](#)” Section concludes by summarizing the critical findings and implications.

Methodology

The hybrid optimization of multiple energy resources (HOMER) software developed by the National Renewable Energy Laboratory (NREL) was applied to analyze the techno-economic feasibility of the optimized hybrid energy systems. The HOMER plans and evaluates small-scale power microgrid systems, such as RESs, traditional DG, and energy storage. Additionally, the HOMER optimizes the microgrid design and operation, considering numerous factors (cost, performance, and environmental impact). The HOMER also utilizes load profiles, system component inputs, and technical constraints to conduct simulations based on various system configurations. It then provides optimal system configurations based on objective functions, including NPC, LCOE, excess electricity, and unmet load.

The analysis was constrained and implemented to optimize system design configurations. A decision-making strategy was proposed to identify the optimal energy system combinations. These combinations derived from HOMER Pro® were assessed using a multi-objective function that considered the performance parameters of the hybrid energy systems. The performance parameters were reliability, economics, feasibility, and environment. A quantitative approach was applied to measure these parameters and increase objectivity. The suggested optimization approach was then investigated using an optimal design case study (stand-alone HES in Lower Shabelle Afgoye, Somalia). Figure 2 portrays the schematic flowchart for the optimization evaluation. Several energy sources were investigated in the simulations to improve the system setup. Three case scenarios (PV/BESS, PV/DG, and PV/DG/BESS) were compared using performance metrics and HOMER Pro® simulation software.

Description of the site under study

The case study focused on a small Afgoye Town village, the third largest city in Southwest Somalia. Afgoye is located 30 km north of Mogadishu in the lower Shebelle valley (latitude: 2.1426°, longitude: 45.1167°) and is 73.29 m above sea level. The average daily population of this village is 59 households. Subsequently, the peak demand is 23.54 kW, and the daily load is 135.57 kWh/day based on the appliances in each residence.

Afgoye solar resource and temperature

The annual average sunlight amount in Somalia is between 2900 and 3100 h (8–8.5 h daily); the average temperature (27 °C) is not too high for the PV system to function for its lifetime. Moreover, the solar irradiance and air temperature statistics of Afgoye are acquired online by identifying the latitude and longitude of the location using the HOMER Pro® software. In Afgoye, the average daily solar radiation is 5.66 kWh/m²/day, with an annual GHI of 2,450 kWh/m²/day. Figure 3 illustrates the monthly average daily solar radiation ranging from 5 to 6.55 kWh/m²/day. The highest and lowest monthly average air temperatures in Afgoye were recorded in March and August (monthly average of 27.22 °C) at 28.67 and 25.86 °C, respectively (see Fig. 4).

Load profile

In this study, the HES addressed the electrical requirements of community-based households. Rural residential regions produced a lower energy consumption than urban residential districts. Residential lighting, fans, and refrigerators accounted for most of the overall power use of the community. Lighting, fans, and freezers were typical everyday utility demands contributing to considerable power use in the agricultural village. The electric load was meticulously measured based on the households’ power consumption throughout the peak summer season. Thus, the current electrical demand of the villages was supplied using two 15 kW DGs operating for 24 h. This study then investigated the electrical demand and conducted a techno-economic assessment of an HES based on load demand and environmental factors.

Table 2 lists the load consumption-based appliances using the Banadir Electric Company (BECO) data for two years. Notably, 59 households consumed approximately 136.16 kWh over two years for home appliances. The BECO data were used to calculate the peak load using HOMER Pro® software to ensure the system reliability of the average electric load (kWh/day). Subsequently, the analysis recorded baseline and scale values of 116.38 and 136.38 kWh, respectively. Observations indicated that the system peak load was

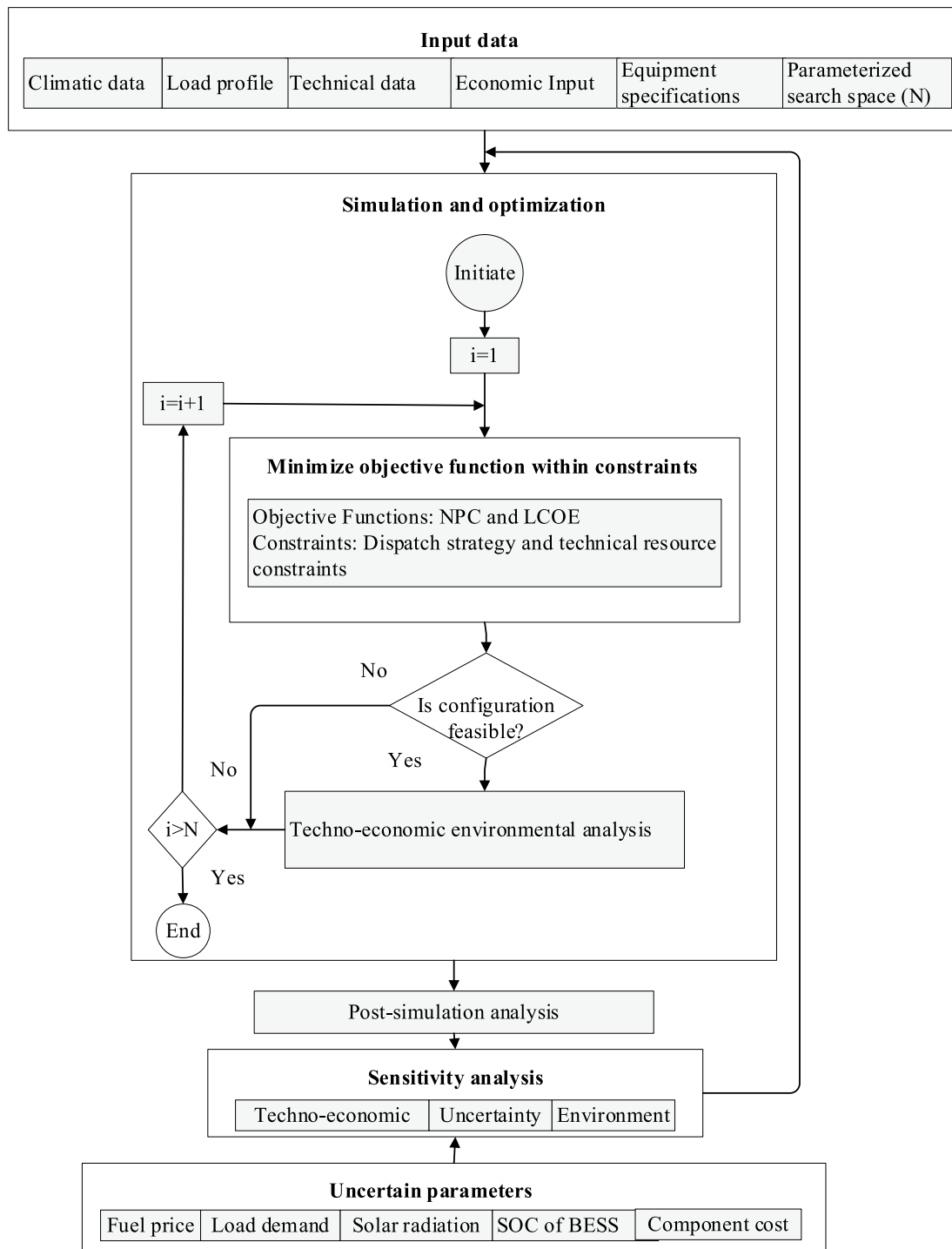


Fig. 2 Schematic flowchart of HES design optimization

23.54 kW, and the primary base load was generated with HOMER Pro® (see Table 3).

Figure 5 exhibits the daily load with monthly, seasonal, and annual profiles of the community over the summer. The daily load profiles revealed the morning and evening demand peaks, particularly for residential households. The

daytime load was typically smaller as most residents were likely at work. Nevertheless, the nighttime load increased when all the residents were at home. Figure 6 depicts the monthly variance generated from the power load demand, with fans constituting most of the load. The average

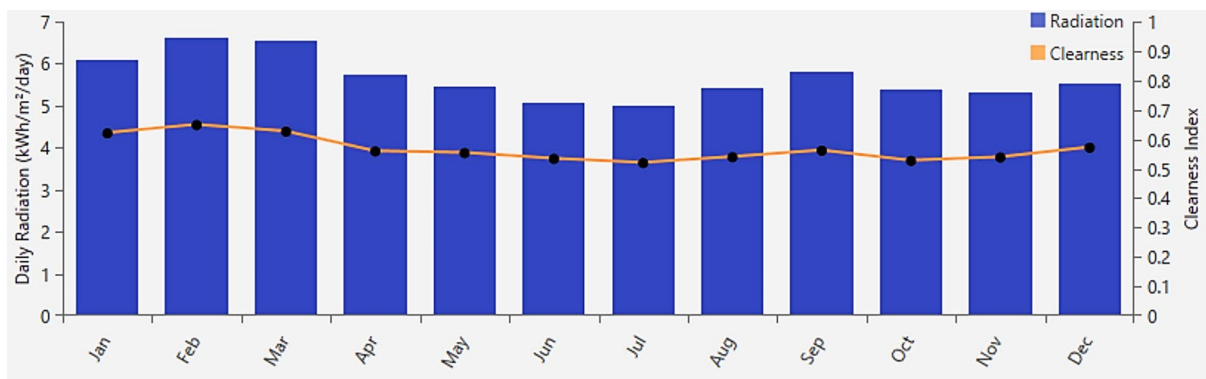


Fig. 3 Monthly average global horizontal irradiance and clearness index for the selected area

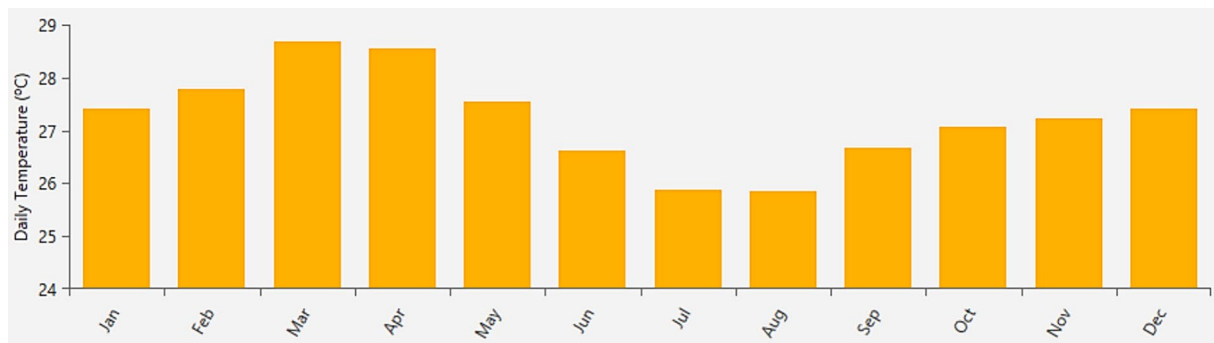


Fig. 4 Monthly average temperature of the selected area

Table 2 Summary of the electrical load power demand for the domestic community

No	Domestic load	Power (watts)	Equipment amount	Usage (hours)	Total load (Wh)
1	CFLs lamps	10	8	6	480
2	Fans	40	3	7	840
3	Washing machine	220	1	1	220
4	Television	60	1	3	180
5	Refrigerator	15	1	24	360
6	Water pump	220	1	1	220
The total load for 01 household					2.3 kWh
The total load for 59 households					135.7 kWh

Table 3 Summary of the scale and baseline electric load metrics

Metric	Baseline	Scaled
Average (kWh/day)	116.38	136.38
Average (kW)	4.85	5.68
Peak (kW)	21.40	23.54
Load factor	0.27	0.27

daily and monthly power consumption in April–May was 85 kWh/day and 2,566.52 kWh/month, respectively.

System component expenses

This study required identifying several variables to reduce the life cycle cost using the HOMER Pro® model in developing a techno-economic analysis of the optimal system. Variable examples included the load, PV panels, battery storage, DG, and converters. The primary components of the investigated energy systems were the DG, PV, BESS,

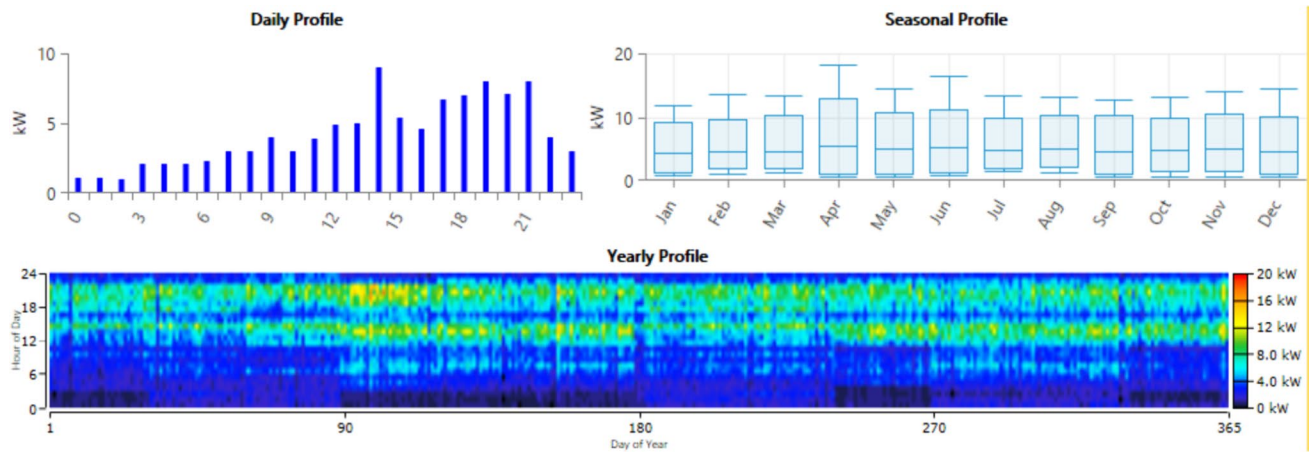


Fig. 5 Daily load, seasonal, and yearly profiles of the load data

and converter. Input factors were required to produce optimal system sizing, such as initial investment, operating, maintenance, and replacement costs, project lifespan, and technical requirements for each component. Table 4 tabulates the required technical details and input settings for each component.

Modeling of the proposed hybrid power system

The PV and BESS in the proposed hybrid power system were regarded as power sources to satisfy requirements using a combined dispatch method. Initially, the PV powered the load when the output exceeded the required energy. Excess energy was then stored in the BESS through a charging procedure. Conversely, the power stored in the BESS supplied the load at the lowest safe battery level when the BESS's state of charge (SOC) exceeded the minimum, and the PV was less than the

load. The backup DG finally supplied the load with electricity when the PV and BESS were insufficient.

Figure 7 reveals the PV/DG/BESS hybrid energy system. The direct current (DC) from the PV array was different from the alternating current (AC) produced by the DG. A DC/AC converter was applied in the BESS to generate DC power and deliver AC by converting DC to AC power. The converter also provided the necessary AC power for the load, while the converter operated in both power flow directions with AC/DC and DC/AC converters. Lastly, the HES assessed the viability of various system configurations.

Mathematical modeling of the system component

Energy system economic model

The economic perspective was crucial in this study due to the primary purpose of the HOMER Pro® software to optimize feasible NPC and LOCE expenses, which maintained the system while producing the optimal design. Generally, the lowest overall cost for a system is calculated by computing the life cycle cost or NPC. This cost is the current value of all expenditures incurred during the project lifespan (Roy et al. 2024). The lowest NPC is calculated as Eq. (1):

$$NPC = \frac{C_{ann,tot}}{CRF(i, T_p)} \quad (1)$$

where $C_{ann,tot}$ is the total annualized cost (\$ per year), i is the annual real interest rate (%), T_p is the project duration, and CRF is the capital recovery factor. Equation (2) is then utilized to determine CRF(i, t) using Eq. (3):

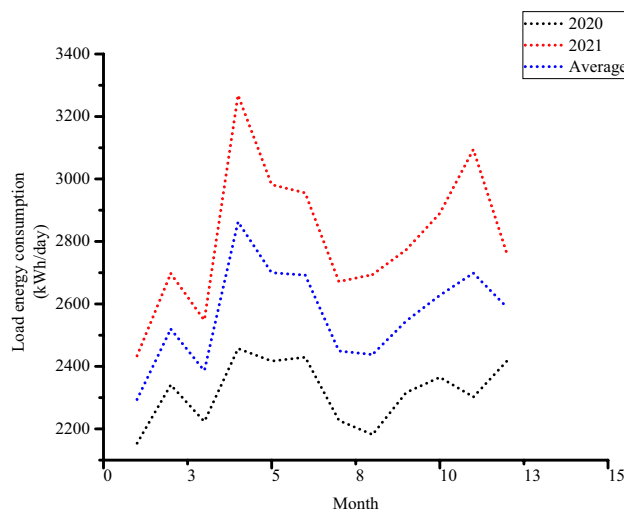


Fig. 6 Monthly energy consumption

Table 4 Summary of key parameters and various components

System components with parameters	Dara description
Diesel generator	
Power capacity	1 kW
Capital cost	\$500 per kW
Replacement cost	\$500 per kW
Operating and maintenance cost	\$0.03 per op. h
Lifetime	15,000 h
PV system	
Power capacity	1 kW
Capital cost	\$480 per kW
Replacement cost	\$480 per kW
Operating and maintenance cost	\$5 per kW/year
Lifetime	25 years
Derating factor (%)	89
Tracking system	Stationary
Temperature effects on power (%/°C)	−0.5
Ground reflection (%)	20
Nominal operating cell temperature (°C)	25
Efficiency at standard test conditions (%)	20.4
Converter	
Power capacity	1 kW
Capital cost	\$200 per kW
Replacement cost	\$200 per kW
Operating and maintenance cost	\$1.5 per kW/year
Lifetime	20 years
Efficiency (%)	95
Relative capacity (%)	100
Battery	
Nominal voltage	6 V
Nominal capacity	1 kWh
Nominal capacity	167 Ah
Roundtrip efficiency (%)	90
Maximum charge current	167 Ah
Maximum discharge current	500 A
Capital cost	\$280
Replacement cost	\$280
Operating and maintenance cost	\$5 per year
Lifetime throughout	336 MWh

$$CRF(i, t) = \frac{i(1+i)^t}{(1+i)^t - 1} \quad (2)$$

$$i = \frac{i' - f}{1 + f} \quad (3)$$

where f , i , and t are the annual inflation rate, nominal interest rate, and total number of years, respectively. When calculating the NPC of a system, the salvage (S) value must be considered. The S value denotes the component value after

it has been used over the project lifespan, which is calculated as Eq. (4) (Anoune et al. 2018):

$$S = C_{\text{rep}} * \left(\frac{T_{\text{rem}}}{T_{\text{comp}}} \right) \quad (4)$$

where C_{rep} is the replacement cost in \$, T_{rem} is the remaining lifetime of the component in years, and T_{comp} is the number of years that the component is expected to last before it fails. The LCOE is calculated using Eq. (5) (Scherzinger and Kaltschmitt 2021):

$$\text{LCOE} = \frac{(AN_C + C_c + C_o)}{\text{Power}_{\text{produced}} - P_{\text{consumed}}} \quad (5)$$

where AN_C is the annual capital costs, C_c is the consumption-based costs, and C_o is the operational costs.

Electrical modeling

Equation (6) presents the renewable fraction (f_{ren}), which is the energy percentage supplied to the load by RE sources as follows (HOMER-ENERGI 2016):

$$f_{\text{ren}} = \left(1 - \frac{E_{\text{non-ren}} + H_{\text{non-ren}}}{E_{\text{served}} + H_{\text{served}}} \right) \times 100\% \quad (6)$$

where $E_{\text{non-ren}}$ is the non-renewable electrical generation in kWh/year, $H_{\text{non-ren}}$ is the non-renewable thermal generation in kWh/year, E_{served} is the yearly electrical consumption in kWh/year, and H_{served} is the total thermal load amount in kWh/year. The total excess electricity fraction (F_{excess}) is the ratio of the total excess electricity to the total electricity production, as shown in Eq. (7) (HOMER-ENERGI 2016):

$$F_{\text{excess}} = \frac{E_{\text{excess}}}{E_{\text{pro}}} \times 100\% \quad (7)$$

where E_{excess} is the total annual excess electricity consumption in kWh/year and E_{pro} is the annual electrical production in kWh/year. The capacity shortage fraction (CSF) is calculated as Eq. (8) (HOMER-ENERGI 2016):

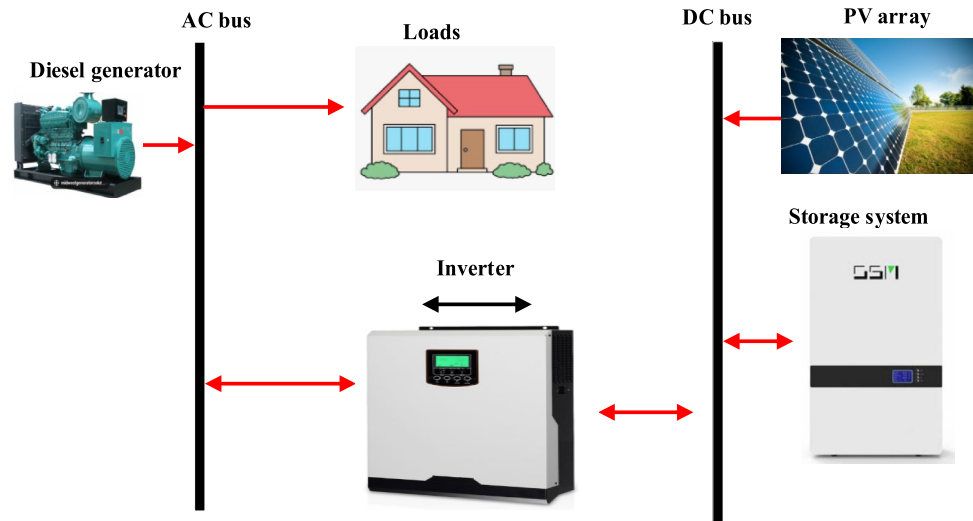
$$\text{CSF} = \frac{E_{\text{cs}}}{E_{\text{required}}} \times 100\% \quad (8)$$

where E_{cs} is the total excess electricity in kWh/year and E_{required} is the annual electricity required in kWh/year.

Modeling of the power output of a PV array

The general PV power output mostly depends on the amount of solar radiation in a selected area. On the contrary, the orientation and tilt angle of the PV panel significantly impact the solar energy quantity used in generating PV power. The

Fig. 7 Schematic representation of the proposed hybrid power system



HOMER computes the PV output power (HOMER-ENERGI 2016) as Eq. (9):

$$P_{PV} = Y_{PV} f_{PV} \left(\frac{G_T}{G_{T,STC}} \right) [1 + \alpha_P (T_C - T_{C,STC})] \quad (9)$$

where P_{PV} is the PV array power output (kW), Y_{PV} is the PV power output under standard test conditions (STC) (kW), α_P is the temperature coefficient of the power ($\%/^{\circ}\text{C}$), f_{PV} is the PV derating factor (%), $G_{T,STC}$ is the incident radiation at STC (1 kW/m^2), G_T is the solar radiation incident on the PV array at the current time step (kW/m^2), T_C is the PV cell temperature ($^{\circ}\text{C}$), and $T_{C,STC}$ is the PV cell temperature at the STC (25°C). The PV cell temperature (T_C) is the module surface temperature, which T_C is equivalent to the ambient temperature at night. During daylight, T_C exceeds the ambient temperature by at least 20°C . An expression is used to calculate T_C as Eq. (10) (HOMER-ENERGI 2016):

$$T_C = T_a + (T_{C,NOCT} - T_{a,NOCT}) \left(\frac{G_T}{G_{T,NOCT}} \right) \left(1 - \frac{\eta_{mp}}{\tau\alpha} \right) \quad (10)$$

where T_a is the ambient air temperature, $T_{C,NOCT}$ is the nominal operating cell temperature, $T_{a,NOCT}$ is the ambient air temperature (NOCT is denoted as 20°C), $G_{T,NOCT}$ is the solar radiation (NOCT is defined as 1 kW/m^2), η_{mp} is the maximum power point efficiency under STC (%), and $\tau\alpha$ is the temperature coefficient of power ($\%/^{\circ}\text{C}$). Nevertheless, T_C influences the η_{mp} efficiency. Assuming that the relationship between efficiency and temperature is linear, Eq. (11) can be expressed as follows (HOMER-ENERGI 2016):

$$\eta_{mp} = \eta_{mp,STC} [1 + \alpha_P (T_C - T_{C,STC})] \quad (11)$$

where $\eta_{mp,STC}$ is the maximum power point efficiency under STC (%), α_P is the temperature coefficient of power ($\%/^{\circ}\text{C}$),

and $T_{C,STC}$ is the cell temperature under STC (25°C). Typically, the temperature coefficient of power is negative, indicating a decline in PV array efficiency as cell temperature rises. Therefore, cell temperature is determined by incorporating the efficiency equation into the previously stated equation for cell temperature, as shown in Eq. (12) (HOMER-ENERGI 2016):

$$T_C = \frac{T_a + (T_{C,NOCT} - T_{a,NOCT}) \left(\frac{G_T}{G_{T,NOCT}} \right) \left[1 - \frac{\eta_{mp,STC} (1 - \alpha_P T_{C,STC})}{\tau\alpha} \right]}{1 + (T_{C,NOCT} - T_{a,NOCT}) \left(\frac{G_T}{G_{T,NOCT}} \right) \left(\frac{\alpha_P \eta_{mp,STC}}{\tau\alpha} \right)} \quad (12)$$

Battery modeling

Equations (13–16) present a realistic calculation of the maximum energy amount stored in a battery, as follows (HOMER-ENERGI 2016):

$$P_{\text{Batt, Cmax}} = \frac{\text{MIN}(P_{\text{Batt, Cmax, Kbm}}, P_{\text{Batt, Cmax, mcr}}, P_{\text{Batt, Cmax, mcc}})}{\eta_{\text{Batt, C}}} \quad (13)$$

$P_{\text{Batt, Cmax}}$ (kW) is the maximum amount of power that the battery system can absorb. $P_{\text{Batt, Cmax, Kbm}}$ (kW) the maximum power can be stored based on the kinetic battery model. $P_{\text{Batt, Cmax, mcr}}$ (kW) is the storage charge power corresponding to the maximum charge rate. $P_{\text{Batt, Cmax, mcc}}$ (kW) is the storage charge power corresponding to the maximum charge current. $\eta_{\text{Batt, C}}$ is the storage charge efficiency. The overall storage capacity is determined by the minimum of these three amounts, which can be computed using the following approach:

$$P_{\text{Batt, Cmax, kbm}} = \frac{kQ_1 e^{-k\Delta t} + Q_{kc}(1 + e^{-k\Delta t})}{1 - e^{-k\Delta t} + c(k\Delta t - 1 + e^{-k\Delta t})} \quad (14)$$

$$P_{\text{Batt, Cmax, mcr}} = \frac{(1 - e^{-\alpha\Delta t})(Q_{\max} - Q)}{\Delta t} \quad (15)$$

$$P_{\text{Batt, Cmax, mcc}} = \frac{N_{\text{Batt}} \times I_{\max} \times V_{\text{nom}}}{1000} \quad (16)$$

where k is the storage constant rate (h^{-1}), Q_1 is the available battery energy at the initial time step (kWh), Δt is the time step duration (hour), Q is the battery energy available during the initial time step (kWh), c is the battery capacity ratio, α_c is the battery's maximum charge rate (A/Ah), $Q_{\max}$ is the total storage capacity (kWh), N_{Batt} is the number of batteries, $I_{\max}$ is the battery maximum charging current (A), and V_{nom} is the battery's nominal voltage.

Modeling of diesel generator

When the generator operates at a given time step, HOMER applies Eq. (17) to calculate the fuel consumption rate, as follows (HOMER-ENERGI 2016):

$$F = (F_{0,\text{gen}} \cdot Y_{\text{gen}}) + (F_{1,\text{gen}} \cdot P_{\text{gen}}) \quad (17)$$

where F is the fuel consumption (units/h) at the current time step, $F_{0,\text{gen}}$ is the generator fuel curve intercept coefficient (units/h/kW), Y_{gen} is the rated generator capacity (kW), $F_{1,\text{gen}}$ is the fuel curve slope for the generator (units/h/kW), and P_{gen} is the electric generator output (kW).

Modeling of power converter unit

The power converter allows bidirectional power flow between DC and AC while functioning as an inverter to convert solar DC power to load-required AC power. The power converter is also a rectifier that transforms the DG-produced AC power into DC power for battery charging. A converter is typically chosen based on the power utilized by the solar module. This study recorded a 95% converter efficiency. The power capacity is determined using Eq. (18) as follows:

$$\eta_{\text{cnv}} = \frac{P_{\text{output}}}{P_{\text{input}}} \quad (18)$$

where η_{cnv} is the converter efficiency, P_{input} is the power input arising out of the converter and P_{output} is the power output arising out of the converter.

System dispatch strategy

Dispatch strategies regulate energy source applications and the battery charging and discharging rates in different scenarios. Several dispatch strategies for RE-based hybrid power systems have been explored in multiple studies (Javed et al. 2020; Chaurasia, Gairola, and Pal 2022). Interestingly, cycle charging (CC) and load following (LF) were discussed in a study by Aziz As et al. (Aziz et al. 2019). The primary role of the generator in the CC strategy is to supply the primary load. Once the primary demand is satisfied, the excess energy charges the battery. Nonetheless, the LF dispatch strategy only generates enough power to provide the primary load whenever a generator is activated—the RE sources power low-priority objectives, including charging storage banks or deferred loads. Consequently, applying DG effectively in the LF method increases the hybrid system's performance over that of CC. This study employed the LF dispatch control strategy to verify the efficacy and performance of the system (see Fig. 8).

Sensitivity analysis

The input parameter influenced the performance of the system and was evaluated using a sensitivity analysis. Generally, the HOMER Pro® software offers system designers feasibility and economic indicators to verify system feasibility and reduce NPC and operating costs depending on the sensitivity of the input parameters. This study applied the HOMER Pro® software to simulate the system configuration, considering the community load and 10 to 26 kW DG with 10–25 kWp PV array capacity. The 25-year operating cost of the system was determined by factoring in various uncertainty parameters throughout the simulation and optimization stages. HOMER Pro® handles parameters like local load variations, inflation rates, nominal discount rates, solar irradiation levels, and prices of PV systems, batteries, and diesel. Moreover, the model illustrated the sensitivity impact on the NPC of the optimal system. In this study, baseline rates for global fuel price, solar radiation, minimum state of charge of BESS (%), and community load demand were \$0.50/L, 5.66 kWh/m²/day, 20%, and 136.38 kWh/day, respectively. The sensitivity parameters explored are detailed in Table 5. The proposed HES is expected to last 25 years. A techno-economic feasibility assessment determined the optimal energy technology mix for different HES configurations.

Results and discussions

This study simulated the load profile and available resources in the specified area. The HOMER Pro® simulation tool determined the optimal system based on economic factors.

Fig. 8 Diagram adopting the LF-based strategy for meeting load demand

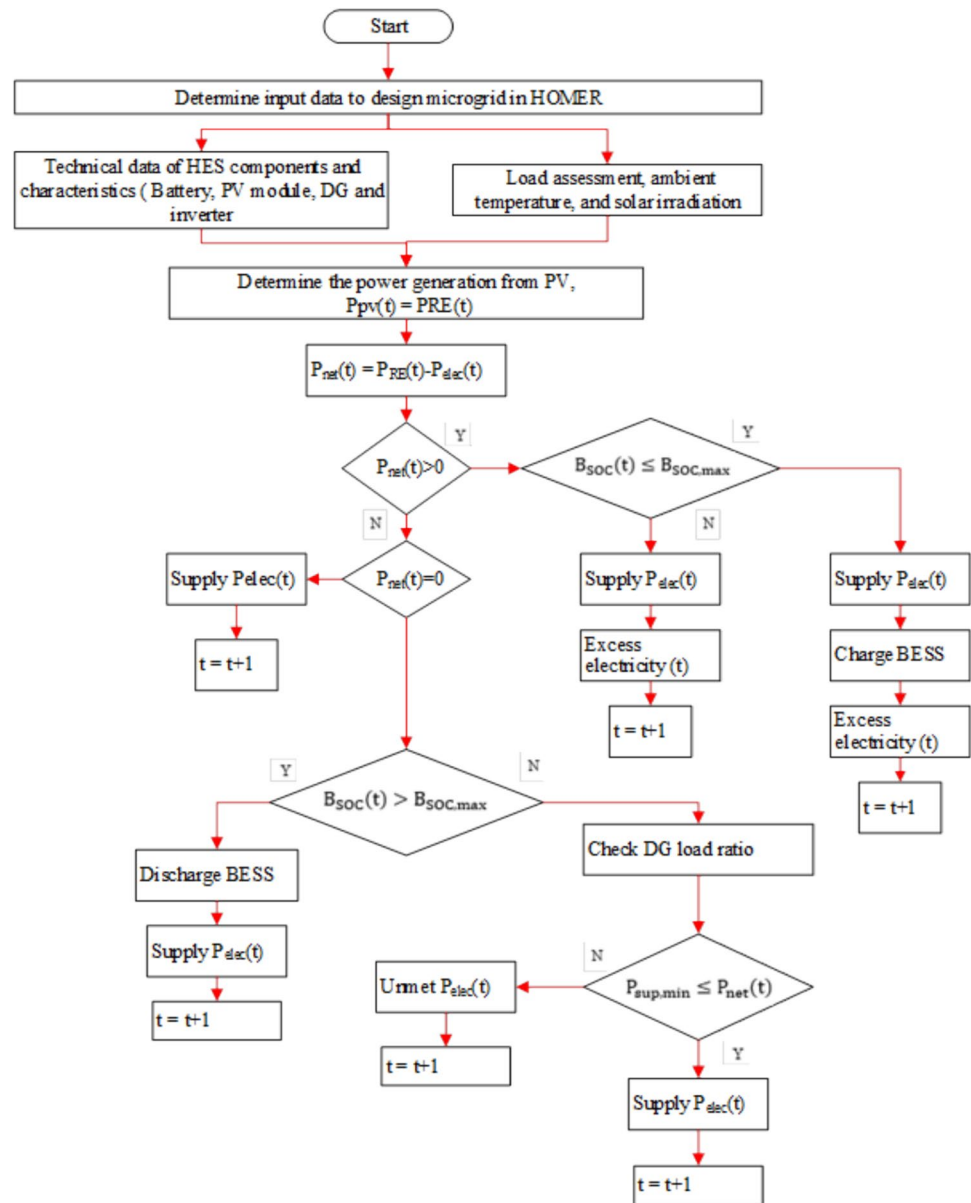


Table 5 Sensitivity parameters

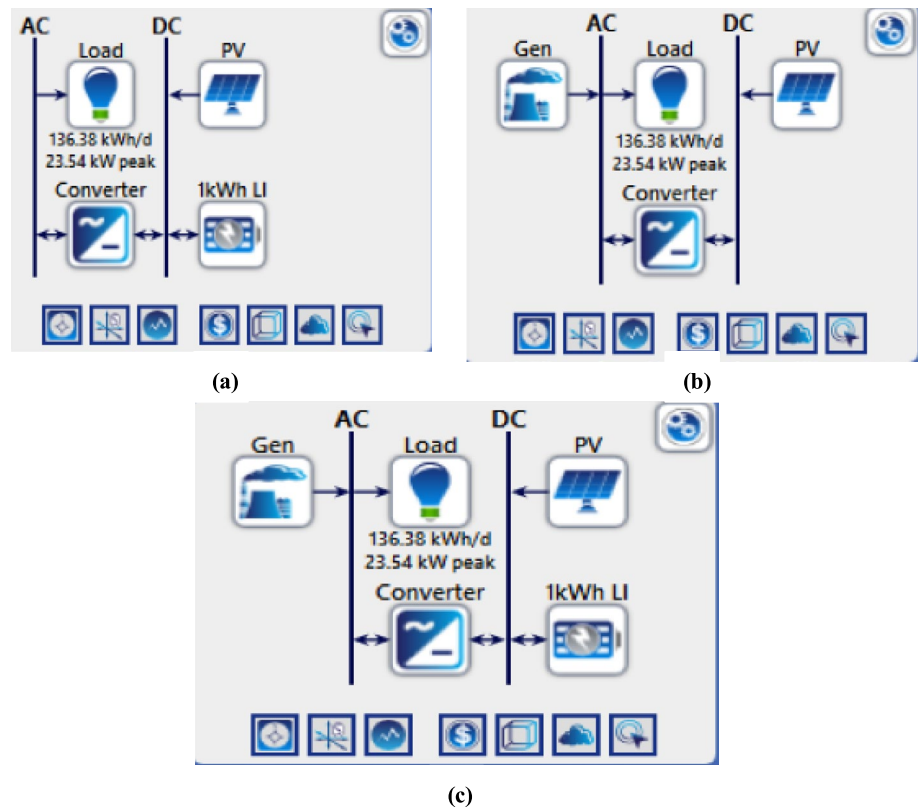
Pattern	Sensitivity quantities					
Solar radiation (kWh/m ² /day)	5.0	5.33	5.66	5.99	6.32	6.65
Minimum SOC of BESS (%)	20	25	30	35	40	45
Diesel fuel price (\$/L)	0.5	0.675	0.85	1.025	1.20	1.375
Load demand (kWh/day)	130	133.19	136.38	139.57	142.76	145.95

The technical data for each hybrid system component (PV, DG, BESS, and converter) was assessed based on the temperature data, load profile data, and real-time PV irradiance at the specified site. Several adoptions and limits were considered for modeling and simulating the HESs (see Fig. 9). The electrical and economic factors with eco-friendly performance were also investigated.

Comparative analysis of hybrid power system

This study examined various production solutions to meet the existing and future load demands for the Afgoye community. Three configurations (PV/BESS, PV/DG, and PV/DG/BESS) were studied by combining relevant RE generation technologies to obtain the most appropriate configuration

Fig. 9 Diagram indicating the hybrid system configuration comparison: **a** PV/BESS, **b** PV/DG, and **c** PV/DG/BESS



regarding technological, economic, and environmental factors (see Fig. 9). The three configurations were compared to establish the optimal design for remote rural locations from the simulation parameters and the HOMER Pro® study outcome. These DG/PV, PV/BESS, and PV/DG/BESS system performances were assessed in relation to the optimal system and sensitivity studies. Compared to other configurations, the PV/DG/BESS system (Configuration 3) was the most optimal system based on the performance metrics. The optimal model parameters were 25 kWp, 26 kW, and 112 kWh for the PV, DG, and BESS, respectively. A 15 kW inverter was also recorded for a 49,779 kWh/year load. The following subsections depict detailed comparisons of energy systems.

Stand-alone PV/BESS system

This subsection described 100% RE in Configuration 1 (PV/BESS), achieving the load requirement with the lowest carbon footprint (see Fig. 9a). The fraction limit conditions for the capacity shortage ranged from 0 to 45% with a 5% interval. A PV system with a power output of 25 kWp and a BESS of 94 kWh was necessary to achieve a 0% capacity shortage fraction. In contrast, the LCOE was \$0.095/kWh, and the initial investment and annualized operating cost dropped by \$33,760 and 1216, respectively. Considering sufficient space was available for the installation, the excess

electricity and unmet electric loads were 3.65% and 25.1%, respectively. Meanwhile, reserve power was utilized to solve transient power shortages. The annual power supply capacity was 69.9%, while the capacity shortage fraction was 30.1%. Although the load demand was approximately two to three times higher (midnight through midday), a power blackout was required to be maintained. The technology did not meet power system reliability requirements even if the system was economically cost-effective and environmentally benign. Table 5 summarizes the technical and economic properties of the system with the energy and annual productions.

Based on the International Energy Agency (IEA) and Asian Development Bank, the Emission Factor (EF) was evaluated to reduce CO₂ emissions in PV projects when the grid EF was considered ("International Energy Agency Emission Factors 2020" 2020; ADB 2017). Equations 20 and 21 demonstrate that the PV system produces 42 MWh/year of power, hence emitting 1701.24 kg CO₂eq as a baseline emission (BE) as follows:

$$EE_{\text{grid}} = \frac{\text{BE of CO}_2(\text{kg or tons})}{\text{Annual electricity production} \left(\frac{\text{MWh}}{\text{Year}} \right)} \quad (19)$$

$$\text{Emission Reduction (ER)} = \text{BE} - \text{PE} \quad (20)$$

The BE formula is obtained by multiplying the energy production from RE projects (REP) with the combined EF of gCO₂eq per MWh or equivalent as follows (see Eq. 22) (IFI TWG 2019):

$$BE = \text{energy output(REP)} * EF \quad (21)$$

The life cycle carbon footprint of a PV system was approximately 40 g CO₂eq/kWh (NREL 2012). This outcome is summarized as follows:

1. Approximately 60 to 70% of the carbon effect was attributable to the upstream plant processes, which included modules, inverters, supports, and other components.
2. Approximately 21 to 26% was contributed by operational activities.
3. Approximately 5 to 20% was associated with panels' disposal or downstream processes.

The carbon footprint amount of the project or Project Emission (PE) was approximately 1701.24 kg CO₂eq emissions annually for 42,531 kWh or 42 MWh of energy output from a PV system.

Hybrid PV/DG system

This subsection described Configuration 2 (hybrid PV/DG), which was optimized to provide the lowest NPC and LCOE performance metrics. The PV system capacity was modified to 25 kWp, and a standby DG of 25 kW was used. Consequently, Configuration 2 generated an NPC of \$369,317 and an LCOE of \$0.513/kW. The excess electricity and RF were 57.9 and 0%, respectively. Furthermore, the capacity shortage fraction and unmet electric load were 0%. Table 6 depicts all optimized parameters and an overview of the system configuration costs. Configuration 2 produced the highest system costs and technical considerations in NPC,

LCOE, and excess electricity. The estimated operating, fuel, and salvage costs were \$23,600 (annually), \$13,495, and \$1,101, respectively.

The scheme study exhibited that the DG functioned for 5,952 h yearly, burning 26,990 L of fuel per year with a 347 L/kWh rate. This finding resulted in an additional fuel expenditure of \$195,001 during the project. Consequently, this technology was unsuitable for isolated areas due to its harmful economic and environmental effects. Limited resources and infrastructure posed financial issues, while the potential ecological impact of the system compromised environmental sustainability. Thus, considering these aspects was essential to ensure viability and eco-friendly solutions for remote locations.

Hybrid PV/DG/BESS system

The performance metrics of Configuration 3 (PV/DG/BESS) system were discussed, such as NPC, LCOE, excess electricity, and RF (see Fig. 9c). This system computed NPC, LCOE, excess electricity, and RF values of \$129,679, \$0.180/kWh, 2.27%, and 73%, respectively. A 0% capacity shortage fraction and an unmet electric load were also observed. The annualized operating, gasoline, and salvage expenses were \$4,770, 2,336, and 4,127, respectively (see Table 6). Based on a 0.348 L/kWh consumption rate and a 4,672 L yearly fuel consumption, the fuel expenditures for the 5,952 DG annual operation hours amounted to \$33,757. Moreover, the yearly fuel consumption was 4,672 L or 0.340 L per kWh.

Figure 10 portrays the cumulative nominal cash flow of the PV/BESS, PV/DG, and PV/DG/BESS systems. The PV/BESS acquired the smallest cost (\$51,331) without system reliability, whereas the PV/DG system possessed the highest price (\$369,317) with annualized operating costs to maintain power system reliability. Intriguingly, PV/DG/BESS (Configuration 3) offered the most cost-effective system at

Table 6 Comparison between the technical and economic aspects of several HESs

Systems	PV/BESS	PV/DG	Proposed PV/DG/BESS
Electric production (kWh/year)	42,531	120,5521	55,969
Energy consumption (kWh/year)	37,293	49,779	49,779
Excess electricity (kWh/year)	1,553 (3.65%)	69,594 (57.9%)	1,268 (2.27%)
Unmet electric load (kWh/year)	12,486 (25.1%)	–	0.00
Capacity shortage (kWh/year)	14,982 (30.1)	–	0.00
Renewable fraction (%)	100	0.00	73
Total fuel consumed (L)/year	–	26,990	4,672
Initial capital (\$)	33,760	28,300	60,759
NPC (\$)	51,331	369,317	129,679
LCOE (\$)	0.095	0.513	0.180
Annualized operating cost (\$)	1,216	23,600	4,770
Salvage (\$)	(2,319)	(1,101)	(4,127)

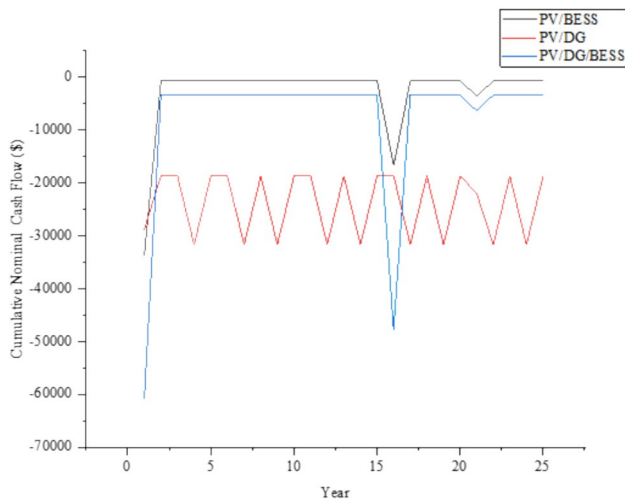


Fig. 10 Cumulative nominal cash flow of PV/BESS and PV/DG/BESS systems

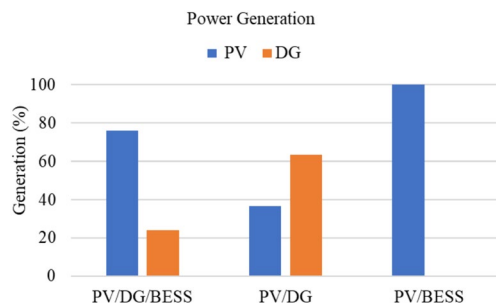


Fig. 11 Power percentage is generated using different energy sources

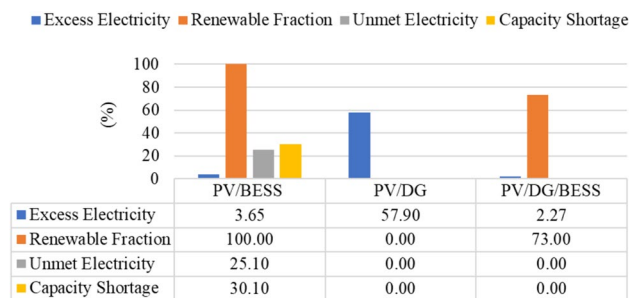


Fig. 12 Excess and unmet electricity, renewable fraction, and capacity shortage

\$129,679. Figure 11 illustrates the power generation from multiple energy sources for the configuration analysis. Configuration 3 comprised the ideal PV/DG/BESS system, in which PV created 76% of the total electricity while DG generated 24%. In Configuration 2, DG and PV produced 64.6% and 35.4% of the total electricity, respectively.

Figure 12 reveals the PV/DG/BESS system (Configuration 3) acquiring 73% of the renewable-based PV power, with a 2.27% maximum excess electricity due to high PV generation. This maximum value was below the commonly accepted 5% threshold. Therefore, hybrid energy-based generation was possible for the Afgoye community considering economic and environmental effects. Configuration 1 produced excess electricity and renewable fractions of 3.65% and 100%, respectively. Likewise, Configuration 2 computed 57.9% and 0%, respectively. In addition, the unmet electric load and capacity shortage were negligible for all configurations. Thus, Configuration 3 could meet the load requirements of the community.

Most expenses were related to DG and BESS, depicted in the NPC for the examined configurations (see Fig. 13). BESS has been operational for 14 years with a 15-year expected lifespan. Although the project lifetime was 25 years, the BESS must be replaced after 10 years. The BESS replacement costs were necessary for all examined configurations, which were the highest over the entire project duration. The fuel needed to operate the DG was also a high-cost factor. Although the optimized PV/DG/BESS system possessed the lowest fuel cost, the PV/DG system demonstrated the highest fuel cost (63.5% of the total electricity).

This study encouraged RE to boost electricity access and minimize greenhouse gas emissions for ecologically friendly and future power systems. Table 6 tabulates a comparative emission parameter analysis for the three analyzed configurations. The PV/BESS system emitted fewer pollutants due to its exclusive RE application. Conversely, the system reliability around midnight diminished significantly. Therefore, the optimized PV/DG/BESS system satisfied all the analyzed criteria, with its configuration emitting 82.7% less pollution than the PV/DG system.

Hybrid energy system optimization results

Table 7 displays the optimal system simulation results. A feasible system configuration was assessed by determining whether the model system performed more efficiently from an economic perspective and power system reliability. The sensitivity analysis reported an LCOE range from \$0.095 to 0.513 per kWh and an energy shortage from 0 to 30.1% for production capacity. Similarly, the PV/DG/BESS system was a viable design with a 0% capacity shortage, fuel consumption of 4,672 L/year, and CO₂ emissions of 12,230 kg/year (see Table 8). Nonetheless, the HES (Configuration 3) acquired a 0% capacity shortage fraction, an unmet electric load of 0%, and 12,330 kg/year of CO₂ emission, less than other systems. The LCOE of \$0.180/kWh indicated this was the most practical configuration. The PV/DG/BESS was the most viable option with the highest performance for serving the selected remote area



Fig. 13 Component and operating costs for PV/BESS, PV/DG, and PV/DG/BESS systems

Table 7 Summary of the comparative emission analysis of the studied configurations

Pollutant emissions (kg/year)	PV/BESS	PV/DG	Proposed PV/DG/BESS
Carbon Dioxide	1701.24	70,648	12,230
Carbon Monoxide	0	445	77.10
Unburned Hydrocarbons	0	19.40	3.360
Sulfur Dioxide	0	173	29.900
Nitrogen Oxides	0	418	72.400
Particulate Matter	0	2.70	0.467

with excess electricity of 1,268 kWh/year (equivalent to 2.27%). This outcome was the most reliable and cost-effective power system, which could be implemented affordably and sustainably to improve quality of life and facilitate electricity recovery. Table 9 presents the annualized costs, including operating and maintenance (O&M) expenses, fuel costs, and replacement costs, for the optimum PV/DG/BESS HES. Table 8 also outlines the total annualized cost for the system.

Table 8 Summary of results for the optimum system

PV (kW _p)	DG (kW)	BESS (kWh)	Converter (kW)	Capital cost (\$)	NPC (\$)	Annualized operating cost (\$/year)	LCOE (\$)	Excess electricity fraction (kWh/year)	RF (%)	CO ₂ (kg/year)	Fuel consumption (L/year)
25	26	112	15.0	60,759	129,679	4,770	0.180	1,268 (2.27%)	73.0	12,230	4,672
25	26	–	16.5	28,300	369,317	23,600	0.513	69,594 (57.90%)	35.4	70,648	26,990
25	–	94	14.5	33,760	51,331	1,216	0.0953	1,553 (3.65%)	100	1,701.24	0.00

Performance assessment of the optimum hybrid energy system

Technical evaluation of optimum system

Figure 9c presents the proposed HES schematic, assessed for load requirements using HOMER Pro®. Alternatively, Fig. 14 demonstrates the monthly electricity generated, power generation by each element of the optimum system and the energy consumption proportion of the AC primary load. The PV generated approximately 76% of the total electricity production of the hybrid system due to the high irradiation potential at this location. Meanwhile, DG was the second-largest energy producer, producing 24% of all energy. Although conventional energy sources were intermittent, a hybrid system mainly relied on a 73% renewable fraction of the PV power source. This system managed excess electricity during power generation and ensured maximum power delivery to the load. Excess electricity was produced at a rate of 1,268 kWh/year, or 2.27% of the total power-generating capacity, which is below the generally acceptable range of 5%. This study aimed to reduce excess electricity to less than 5%, a goal considered feasible for power system plants. To address the issue of excess energy, future load growth should be considered, and additional energy storage systems should be installed.

Table 10 details the size, operating hours, electrical output, and load for each optimal hybrid PV/DG/BESS system component. The PV system could operate for the entire year at a peak output of 24.4 kW and 4,380 h/year. Thus, the suggested system produced a high renewable penetration rate of 85.4%, which indicates a significant energy production benefit while being favorable to the environment following the lowered DG running hours. This result facilitated the ongoing effort to enhance clean energy access in Somalia and other developing nations. Figure 15 also demonstrates that the DG consumes 4,672 L of fuel, 12.8 L/day, and 0.533 L/hour to meet the system load requirements.

A time-series analysis examined the generation sources at specific intervals throughout the year. The PV system supported most loads during the summer. Concurrently, the DG provided minimal support (see Fig. 16). In the early morning, DG was required as PV generation was insufficient to meet the total peak demand. Meanwhile, the BESS stored any excess energy generated by the PV system during the day. The inverter then supplied the load with electricity using the BESS stored energy.

A battery bank with 112 kWh nominal capacity produced an operational autonomy of 15.8 h in this study. This autonomy rating indicated the number of hours the BESS supported the load demand without recharging. The daily and monthly averages of the battery SOC suggest that the shortest charging cycles occurred between

Table 9 Annualized costs for the optimum PV/DG/BESS HES

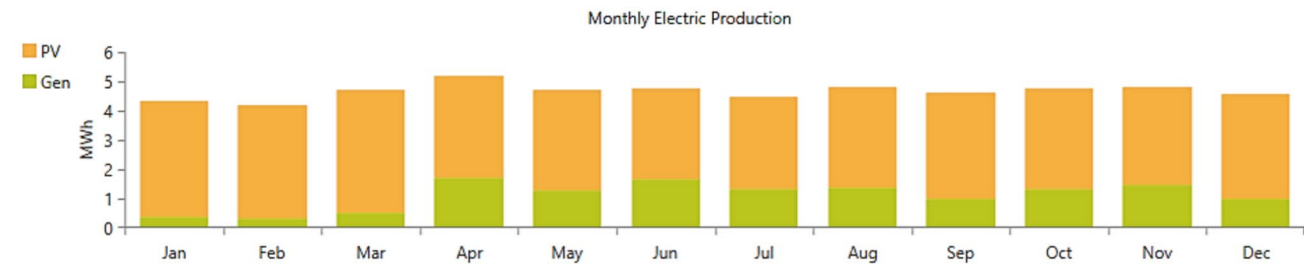
Parameters	Capital (\$)	O&M (\$)	Replacement (\$)	Salvage (\$)	Fuel (\$)	Total (\$)
DG	899.65	808.86	459.61	(76.56)	2,345	4,437
BESS	2,170	0.00	1,131	(148.24)	0.00	3,153
PV	830.45	125.00	0.00	0.00	0.00	955.45
Dispatch strategy	96.89	0.00	0.00	0.00	0.00	96.89
Other	0.00	100.69	0.00	0.00	0.00	100.69
System Converter	207.56	22.49	82.01	(48.76)	0.00	263.29
System	4,205	1,057	1,673	(273.56)	2,345	9,006

Production	kWh/yr	%
Generic flat plate PV	42,531	76.0
Autosize Genset	13,438	24.0
Total	55,969	100

Consumption	kWh/yr	%
AC Primary Load	49,779	100
DC Primary Load	0	0
Deferrable Load	0	0
Total	49,779	100

Quantity	kWh/yr	%
Excess Electricity	1,268	2.27
Unmet Electric Load	0	0
Capacity Shortage	0	0

Quantity	Value	Units
Renewable Fraction	73.0	%
Max. Renew. Penetration	1,342	%

**Fig. 14** Monthly electricity generation, energy generation, and power consumption using optimum configuration**Table 10** Summary of the electrical characteristics of the optimized HES (PV/DG/BESS)

Subunit	Size	Annual hours of operation (hrs)	Maximum power output (kW)	Minimum power output (kW)	Single HES penetration (%)	Capacity factor (%)
PV	25	4,380	24.4	0	85.4	19.4
DG	26	1,033	17.0	13	—	5.90
Converter	15 kW	7912 (Invert); 847 (Rectifier)	13.6 (Inv.); 11.8 (Rec.)	0 (Inv.); 0 (Rec.)	—	32.1 (Inv.); 4.18 (Rec.)
BESS	112 kWh	—	—	—	—	—

April and June (90–180 days) due to cooler temperatures (see Fig. 17). During this period, solar radiation was not excessively high. Thus, the energy stored in the battery increased during the summer months (July through February) when electricity demand was low (270–365 days). Conversely, the overall battery losses through the year are 2,499 kWh/year.

Economic assessments

The optimum PV/DG/BESS HES generated approximately \$129,679 (NPC), while the LCOE and operational costs were \$0.180/kWh and \$4,770 per year, respectively. Figure 18 provides a cost summary of the optimized system. This study concluded that fuel (\$33,756) and initial capital

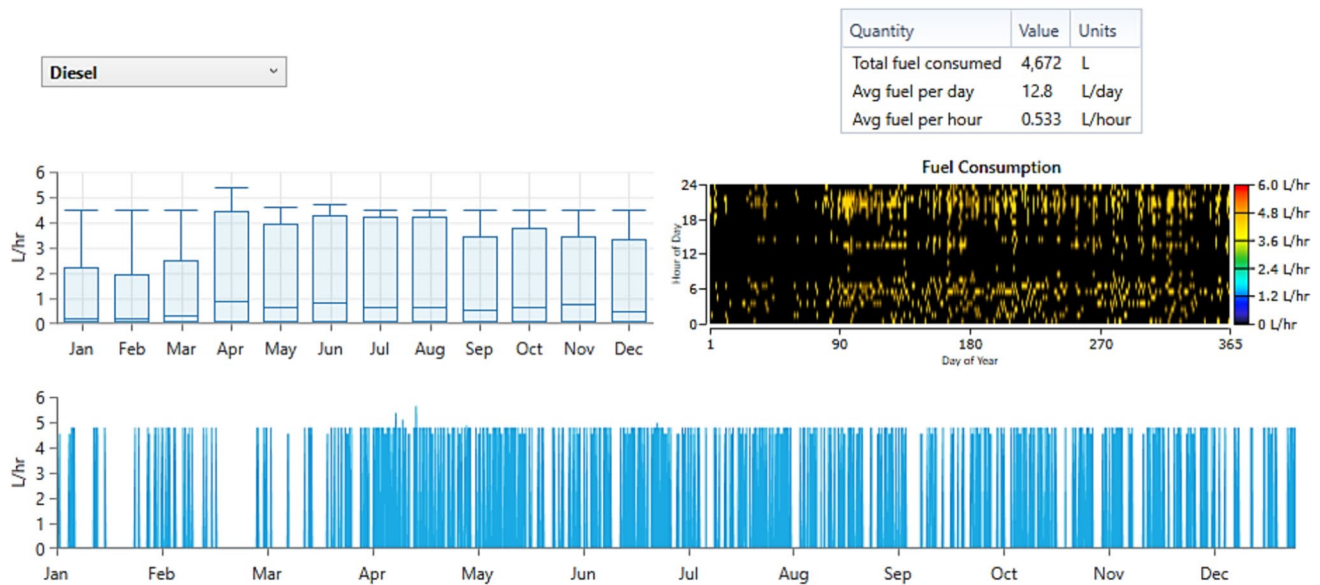


Fig. 15 Annual fuel consumption of the DG

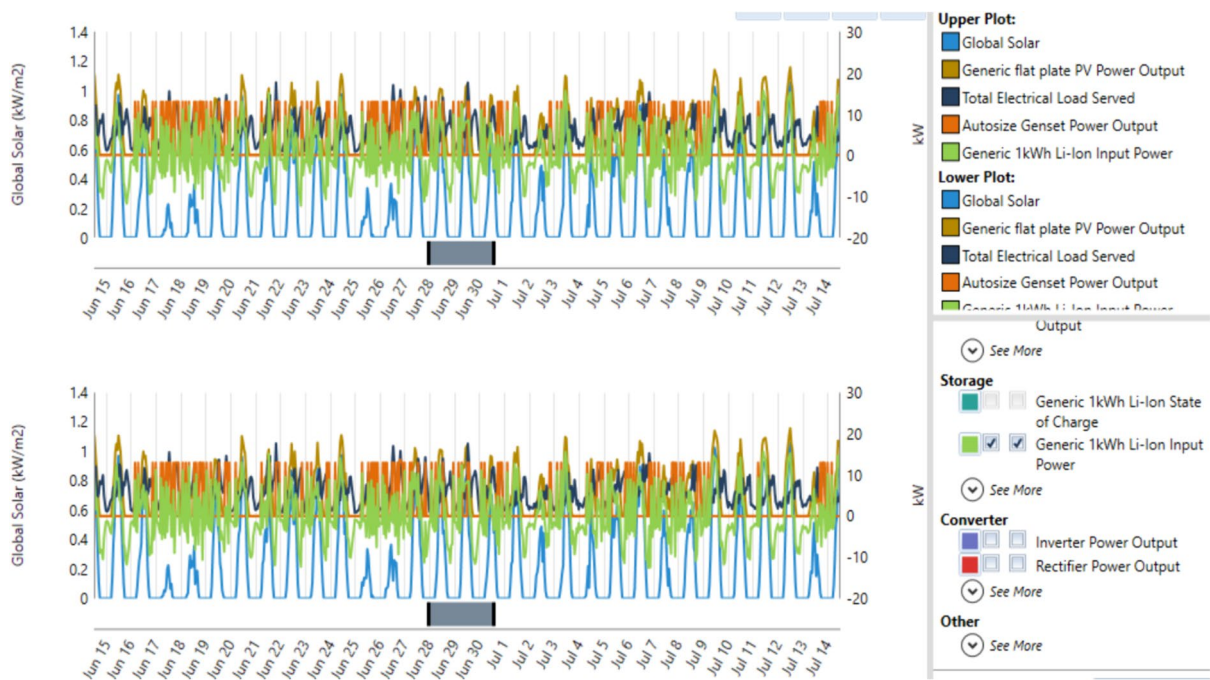
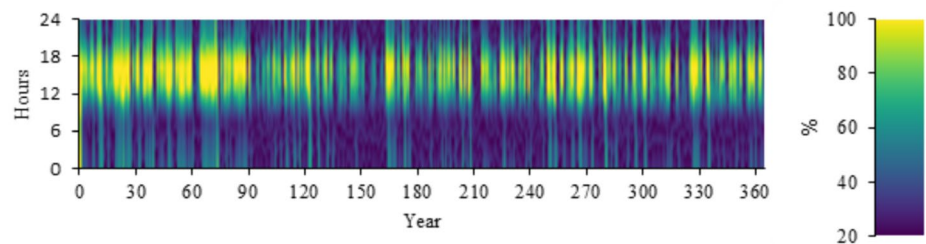


Fig. 16 Energy production from various sources during the summer

Fig. 17 BESS annual state of charge (%)



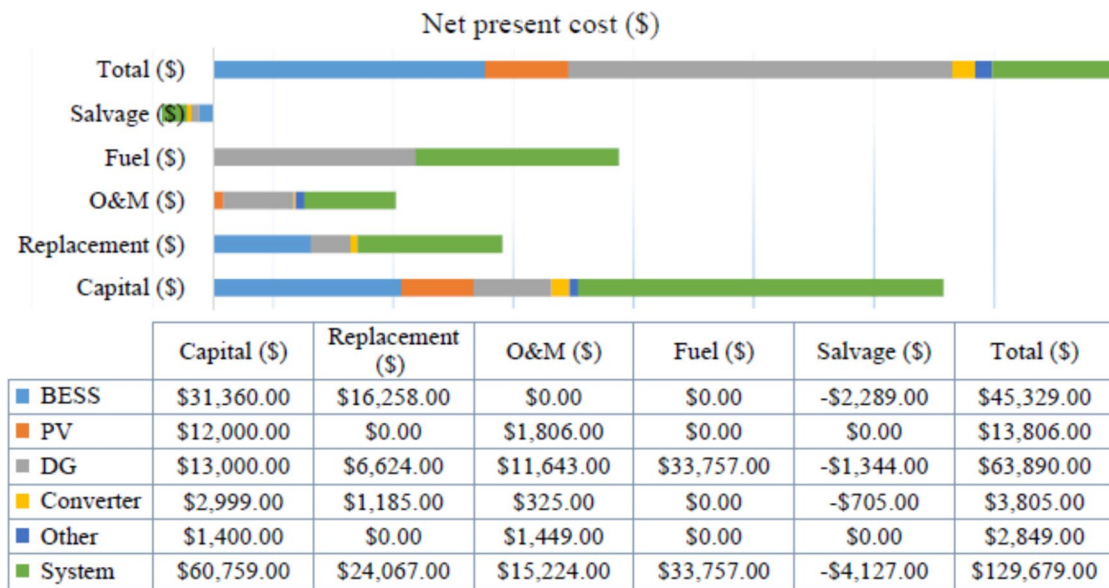
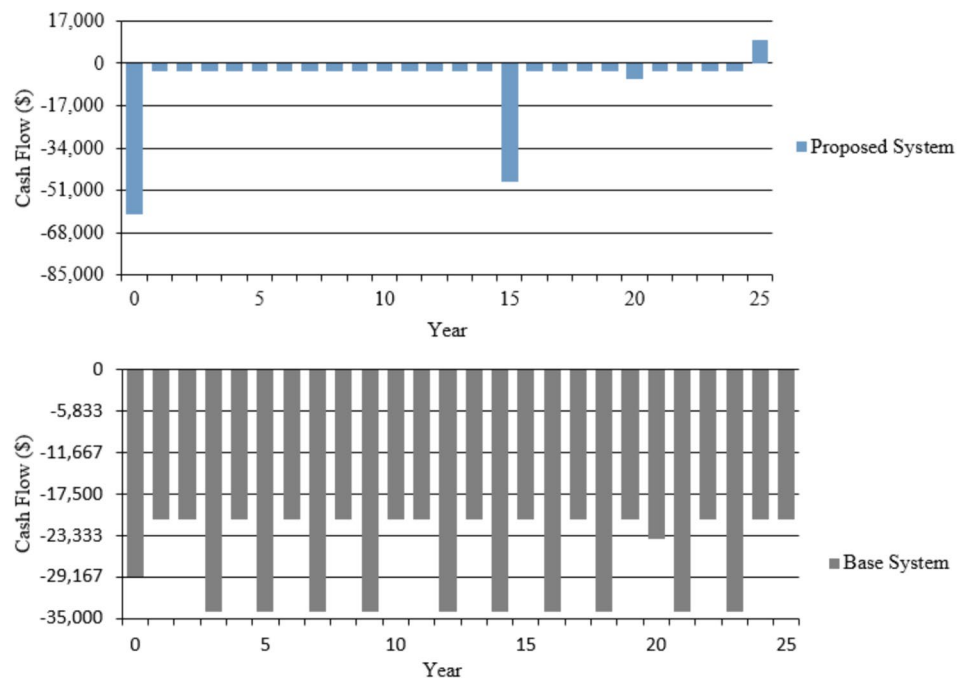


Fig. 18 Cost summary of the optimum hybrid energy system

Fig. 19 Annual nominal cash flows of the proposed and base system



(\$60,756) were the costliest components. The high capital costs and fuel expenses were attributable to the DG working conditions. Furthermore, DG maintenance contributed to the optimal system's high operating and maintenance expenses.

The expected replacement cost was \$24,067 due to the PV system not requiring a replacement. As the component lifespans of the system are from 15 to 25 years, they are comparable to or significantly longer than the overall project duration. The general replacement cost included the BESS,

DG, and converter. Figure 19 exhibits the "nominal cash flow" outcome of the optimal (PV/DG/BESS) and base (DG) systems over a project lifetime of 25 years. The nominal cash flow in the optimal system sustained a constant minimum cost throughout 25 years, whereas a continuous maximum value was maintained in the base system. The hybrid PV/DG/BESS energy system is economically viable in remote locations. Integrating it into the socio-economic context maximizes its potential for sustainable development and

local well-being. This integration fosters economic growth by creating jobs, boosting business activities, and developing supply chains. It improves quality of life with reliable year-round energy for education, healthcare, and living conditions. Additionally, it enhances agricultural productivity and local industries with efficient electric-powered operations. Environmentally, it reduces reliance on fossil fuels, lowering emissions and pollution. Community collaboration strengthens socio-economic formations, as seen in successful projects in developing countries.

Environmental impact assessment

This study considered carbon emissions and other pollutants released during power conversion from fuel combustion, including sulfur dioxide, nitrogen oxide, carbon dioxide, carbon monoxide, particulate matter, and unburned hydrocarbons. Moreover, diesel fuel use was considered in the hazardous emission assessment. The PV/DG/BESS system configuration produced 12,413.23 kg/year of total pollutant emissions, while the PV/DG system configuration generated 80,847.94 kg/year. Figure 20 presents the HOMER Pro® analysis, which demonstrates that the PV/DG/BESS system is both ecologically sustainable and economically viable. The research also significantly lowered NPC and LCOE costs compared to the PV/DG hybrid energy system. Consequently, the optimal system was 84.6% less expensive than the base case, while the proposed system achieved a high renewable penetration rate of 85.4%. This outcome exhibited a considerable benefit for energy production and a favorable influence on reducing DG operating hours and pollutant emissions. Furthermore, the findings of this study significantly contribute to SDG 7 by exhibiting that the PV/DG/BESS system is an affordable, reliable, sustainable, and modern energy solution. The substantial reduction in pollutant emissions and costs, along with the high renewable penetration rate, highlights the system's potential to enhance energy access while mitigating environmental impact. This

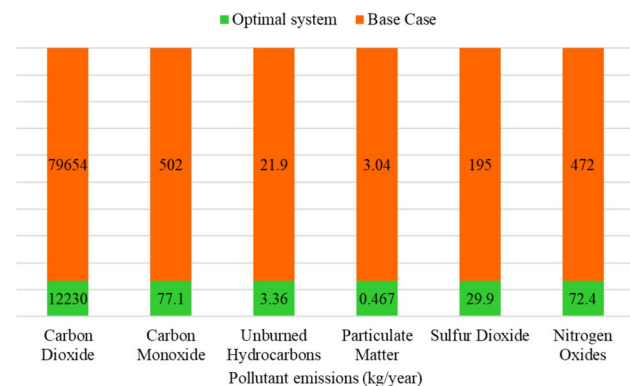


Fig. 20 Greenhouse gas emission comparison for two configurations

makes a strong case for the broader adoption of such HESs to achieve the targets set by the UN.

Sensitivity analysis

In sensitivity analysis, the main parameters affecting the cost of HESs are capital costs covering equipment and installation, operational costs for maintenance and monitoring, load demand, location and site conditions, energy storage requirements, technology efficiency, government policies and incentives, economic factors, and environmental goals. The uncertain parameters selected for this analysis consist of global solar irradiation, minimum state of charge (BESS SOC_{min}), fuel price, and community load demand. When solar irradiance increased, the LCOE of the optimal system reduced, thus indicating that PV/DG with BESS energy generation was more cost-effective than the other configurations. The RE proportion also increased as solar irradiation grew. The impact of fluctuating fuel prices on system costs, comprised of NPC and LCOE, is presented in Fig. 21. Additionally, changes in gasoline prices significantly impact fuel and annualized operating costs but have a marginal impact on O&M costs. Due to an increase in fuel prices from \$0.5 to 0.85 per liter, operational and diesel fuel prices increased by approximately 44.17% and 17.64%, respectively, while

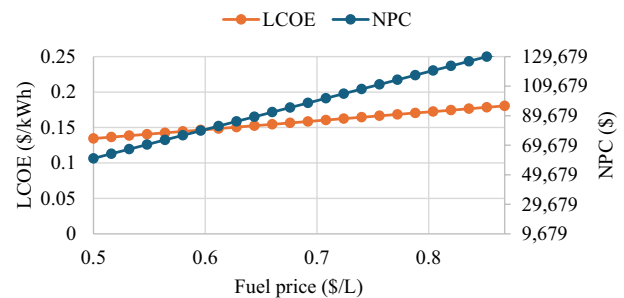


Fig. 21 Effect of variation fuel prices on total NPC and LCOE

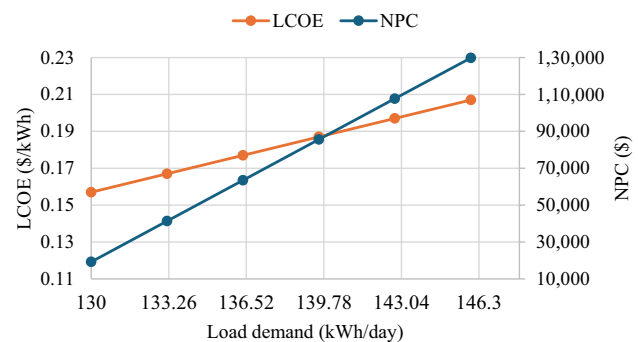


Fig. 22 Impact of load demand growth on total NPC and LCOE

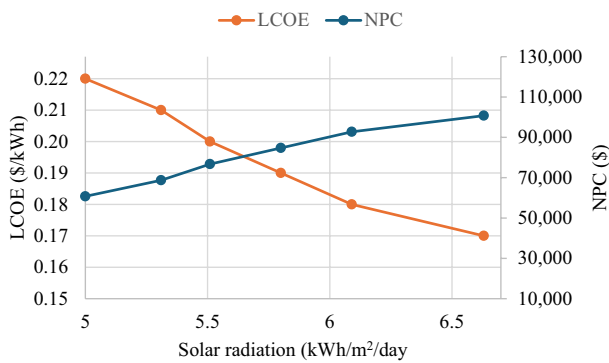


Fig. 23 Impact of fluctuating solar radiation on total NPC and LCOE

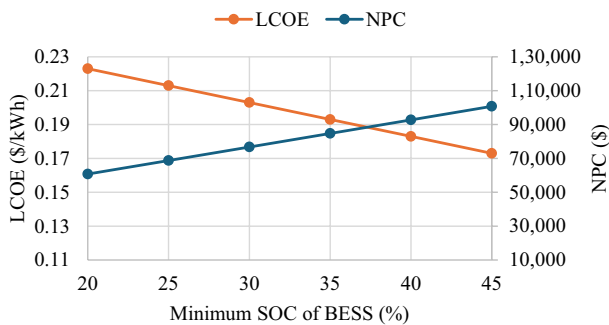


Fig. 24 Impact of fluctuating the BESS SOC_{min} on total NPC and LCOE

the O&M costs increased by 8.19%. This is because of the cost of running the DG when the prices of diesel fuel rise.

Moreover, in all examined sensitivity possibilities, the study exhibits that the NPC of the optimal system configuration increases as both the diesel fuel price and the load growth demand (see Fig. 22). e.g., the NPC rises by approximately 18.57% as fuel prices increase from (\$0.5 to \$0.85)/L and by roughly 20% due to a load demand growth of approximately 3.26 kWh/day compared to the current load consumption. Comparing them, we observe that the LCOE value increases linearly with rising fuel prices.

Figure 23 illustrates the impact of different solar radiation levels on system costs, the renewable fraction used, and the overall fuel consumption. The statistics reveal a reduction in NPC, LCOE, and total fuel utilized of approximately 22.35%, 4.05%, and 9.87%. The share renewable fraction increased by 16.44% as solar radiation at the selected location increased from 5 to 6.63 kWh/m²/day. The decrease in cost parameters can be attributed to reduced fuel consumption, leading to a corresponding decrease in diesel expenses.

In this study, we examined how varying the minimum state of charge (SOC_{min}) of a battery between 20 and 45% impacts the optimal design's performance. We analyzed vital metrics such as system costs and battery SOC_{min}, as shown in

Fig. 24. The findings of the study revealed notable trends. As the SOC_{min} of the battery was increased from 20 to 45%, there was a notable decrease in the values of NPC and LCOE, indicating gains of approximately 22.30% and 3.89%, respectively.

The system denoted the performance uncertainty resulting from significant element fluctuations, while future HES development involved both cost and efficiency concerns. For instance, significant inflation variances occurred due to technical developments, while the PV nominal rate and fuel prices fluctuated over time. Thus, both parameters must be accounted for in fundamental project evaluations. The energy system construction should also be considered in equipment costs, which could vary with time, location, and supplier.

Comparative analysis of the proposed HES versus existing systems

The viability of the proposed system was assessed through a comparative study conducted between the current HES systems built by various researchers over the years and the proposed system. The results of this study are presented in Table 10. It is evident from the study that only the proposed system considers the excess electricity for the development of HES and utilizes it in combination with the BESS to reduce excess electricity and CO₂ emissions while decreasing the LCOE, thereby making the system more sustainable. It is evident that the proposed system has almost a lower LCOE. On the other hand, the proposed system has much lower excess electricity compared to the current system configurations (Jain and Sawle 2021). The HESs in Algeria were evaluated in another Table 11 study, and it was proposed that more RE could be generated by implementing PSO in a PV/diesel/battery configuration. Nevertheless, it is noteworthy that the study does not account for excess electricity, leading to a higher LCOE (\$0.38) compared to the proposed system (Fodhil et al. 2019). The proposed system demonstrated a viable model, boasting a 0% capacity shortage and 4,672 L/year fuel consumption. The proposed system achieved a 0% capacity shortage, 0% unmet electric load, and 12,330 kg/year CO₂ emissions, outperforming other systems with an LCOE of \$0.180/kWh. Furthermore, the PV/DG/BESS emerged as the most practical option, delivering the highest performance with 1,268 kWh/year of excess electricity (2.27%). The proposed optimal system is cost-effective and sustainable, making it an economically viable solution to enhance the quality of life and facilitate electricity recovery.

Conclusions

Island communities face significant difficulties in accessing electricity due to isolation, inadequate infrastructure, and environmental threats. Challenges include unreasonable

Table 11 Comparison of the proposed HES to existing work

Ref	Site	Optimized model	LCOE (\$)	CO ₂ emission (kg/year)	EE (%)	Main features
(Fodhil et al. 2019)	Algeria	PV-DG-BESS	0.38	1994.5	–	The hybrid system met the energy desires of 20 households using PSO
(Cai et al. 2020)	Iran	PV-DG-BESS	0.4429	–	–	A hybrid optimization algorithm finds the optimal capacity, balances load consistency, and minimizes total life cycle cost
(Agyekum and Nutakor 2020)	Ghana	PV-WT-DG-BESS and PV-DG-BESS	0.382 and 0.396	–	–	The techno-economic potential of HESs is evaluated using HOMER
(Mokhtara et al. 2021)	Algeria	PV-DG-BESS	0.21	–	–	The primary objectives were minimizing COE while maximizing system reliability and RF using the PSO algorithm and the ϵ -constraint method
(Jain and Sawle 2021)	India	PV-WT-MH-BT-CT-DG	0.145	13,386	22.4	Assessing town demands implies examining both on-grid and off-grid systems, employing a techno-economic analysis using HOMER
(Menesy et al. 2024)	Saudia Arabia	PV-WT-FC	0.4412	–	–	This study aims to employ three optimization methods (ARO, GWO, WOA) to minimize COE and improve power supply reliability in rural areas
(C. Li et al. 2022a, b, c)	China	PV-BG-BESS	0.24	–	–	Assessments using HOMER software indicated that the PV/BG/battery hybrid system was the most cost-effective option
(Wankouo Ngouieu et al. 2023)	Cameroon	PV-WT-FC and PV-WT-BESS	0.1959 and 0.2386	–	–	A comparative analysis of meta-heuristic optimization methodologies is conducted to develop optimal hybrid PV/WT/FC and PV/WT/BESS systems for continuous power supply to three typical loads in Cameroon
Proposed system	Somalia	PV-DG- BESS	0.180	12,413	2.27	The proposed system assesses the viability of a HES in Somalia, considering various energy sources and costs to find the most suitable solution using HOMER Pro

fuel transportation costs, reliance on imported fuels prone to price changes, and struggles with the accurate assessment of intermittent renewables integration approaches. To overcome these challenges demands adapted approaches such as investing in RE, strengthening infrastructure resilience, and implementing innovative financing methods for sustainable energy access. This paper aims to evaluate the techno-economic feasibility of adopting HESs in isolated communities in Somalia through a case study, with a focus on identifying the optimal system for sustainable rural electrification. The HOMER Pro® optimization tool precisely evaluated the feasibility of the off-grid system, analyzing various configurations, including PV/BESS, PV/DG, and PV/DG/BESS. Moreover, rigorous case study assessments thoroughly evaluated the performance of the proposed system, considering techno-economic factors. The main results of the study are presented as follows:

- 1) The feasibility study explored the opportunities and potential of RE generation implementation in remote rural areas, with a specific focus on Somalia. The aim was to develop a population-beneficial HES influenced by the desires of remote rural communities in Somalia. The assessment demonstrated that the hybrid PV/DG/BESS system was optimal for the community, and the techno-economic and environmental viability study showed that it was economically feasible.
- 2) This proposed system generated the lowest NPC and LCOE of the evaluated system configurations at \$129,679 and \$0.180/kWh, respectively. The payback period, return on investment (ROI), and internal rate of return (IRR) were 1.8 years, 65%, and 66.6%, respectively.
- 3) Interestingly, the proposed system produced the highest RE, renewable penetration rate, and excess electricity at 73%, 85.4%, and 2.27%, respectively. These values were equivalent to 1268 kWh/year and produced 12,413.23 kg/year of total pollutant emissions, below 84.6% of the current power system. The electric load had the most significant impact on the NPC among the six sensitivity factors, which increased as the electrical load demand grew.
- 4) HESs play a crucial role in improving energy access, empowering communities, and boosting public health, thus enhancing overall quality of life. By delivering reliable and affordable energy solutions, especially to underserved areas, HESs enhance vital services like education and healthcare. Community involvement in energy management promotes empowerment and aids the shift to cleaner energy sources, cutting air pollution and improving public health. With a 73% RE fraction, the proposed system offers an accessible solution for developing countries with inadequate power infrastructures. To speed up implementation, targeted incentives for RE investments and community-driven initiatives like energy cooperatives are suggested, driving economic growth and fostering sustainable development.
- 5) It is noteworthy that the implementation faces significant techno-economic challenges due to the variability of RESs like solar, complex integration of multiple energy sources, high upfront costs, ongoing maintenance expenses, limitations of energy storage technologies, fluctuating demand, grid integration issues, regulatory hurdles, and reliance on governmental incentives, policies, political dynamics, and environmental factors. Staying beside technological innovation and market dynamics is crucial for long-term system sustainability and impacts the system's overall NPC.

Future research in this field can assess the effectiveness of integrated systems through the application of optimization techniques. These systems incorporate different energy sources, including solar, wind, geothermal, and conventional energy, alongside energy storage solutions. Subsequent analyses could encompass techno-economic evaluations of power generation. Furthermore, a comprehensive configuration study would be a priority to examine system functionality and interpret the impact of socio-economic, environmental, and governmental policies.

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Author contribution AMS involved in conceptualization, data curation, investigation, methodology, formal analysis, writing—original draft, and writing—review and editing. SM involved in project administration, supervision, validation, funding acquisition, and review and editing. HM involved in supervision, validation, and review and editing. MK involved in writing—review and editing. OA involved in reviewing and editing.

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Data availability The data will be made available upon request.

Declarations

Conflict of interest The authors have no conflicts to disclose.

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